

MTH:3340 ABSTRACT ALGEBRA**QUIZ 1****DUE: FRIDAY JAN. 30**

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)

Your ability to express your solutions is just as important as mathematical accuracy. You are expected to write every solution using complete sentences.

You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

Problem 1. (6 points)

- (a) Find the prime factorizations of 12 and 54.

$$12 = 2^2 \cdot 3 \quad 54 = 2 \cdot 3^3$$

- (b) Find the greatest common divisor and least common multiple of 12 and 54.

$$\gcd(12, 54) = 6 \quad \text{lcm}(12, 54) = 2^2 \cdot 3^3$$

- (c) Find integers r and s such that $\gcd(12, 54) = 12 \cdot r + 54 \cdot s$

$$54 = 4 \cdot 12 + 6$$

$$\text{so} \quad 12(-4) + 54(1) = 6 = \gcd(12, 54)$$

$$\boxed{r = -4, s = 1}$$

Problem 2. (8 points)

(a) Find the prime factorizations of 471 and 562.

$$471 = 3 \cdot 157$$

$$562 = 2 \cdot 281$$

(b) Find the greatest common divisor and least common multiple of 471 and 562.

$$\gcd(471, 562) = 1$$

$$\text{lcm}(471, 562) = 471 \cdot 562$$

(c) Find integers r and s such that $\gcd(471, 562) = 471 \cdot r + 562 \cdot s$

$$\begin{aligned} 562 &= 1 \cdot 471 + 91 \\ 471 &= 5 \cdot 91 + 16 \\ 91 &= 5 \cdot 16 + 11 \\ 16 &= 1 \cdot 11 + 5 \\ 11 &= 2 \cdot 5 + 1 \\ 5 &= 5 \cdot 1 + 0 \\ 5 &= 16 - 11 \\ 11 &= 91 - 5(16) \\ 16 &= 471 - 5 \cdot 91 \\ 91 &= 562 - 471 \end{aligned}$$
$$\begin{aligned} 1 &= 11 - 2(5) \\ 1 &= 11 - 2(16 - 11) \\ 1 &= 3 \cdot 11 - 2 \cdot 16 \\ 1 &= 3(91 - 5 \cdot 16) - 2 \cdot 16 \\ 1 &= 3 \cdot 91 - 17 \cdot 16 \\ 1 &= 3 \cdot 91 - 17(471 - 5 \cdot 91) \\ 1 &= 88 \cdot 91 - 17 \cdot 471 \\ 1 &= 88(562 - 471) - 17 \cdot 471 \\ 1 &= \underline{88} \cdot 562 - \underline{105} \cdot 471 \end{aligned}$$

$s = 88$

$r = -105$

Problem 3. (10 points) Let $A = \{\alpha, \beta, \gamma, \delta\}$, $B = \{1, 2, 3\}$, $C = \{\alpha, 2, \gamma\}$ and list all elements in the following sets.

(a) $C \times B$

$$(\alpha, 1) \quad (\alpha, 2) \quad (\alpha, 3) \quad (2, 1) \quad (2, 2) \quad (2, 3) \\ (\gamma, 1) \quad (\gamma, 2) \quad (\gamma, 3)$$

(b) $A \cap B$

$$A \cap B = \emptyset$$

(c) $A \cap C$

$$\alpha \quad \gamma$$

(d) $B \cup C$

$$1, 2, 3, \alpha, \gamma$$

Problem 4. (5 points) List all elements in the following set.

$$S = \{x \mid x \in \mathbb{N} \text{ and } -5 \leq (x-2)^2 - 6 \leq 3\}.$$

$$-5 \leq (x-2)^2 - 6 < 3$$

$$1 \leq x-2 < 3$$

$$1 \leq (x-2)^2 < 9$$

$$3 \leq x < 5 \text{ gives } x=3, 4$$

$$1 \leq |x-2| < 3$$

$$-1 \geq x-2 > -3$$

$$\text{so } 1 \leq x-2 < 3$$

$$1 \geq x > -1 \text{ gives } x=1 \text{ only}$$

$$\text{or} \\ -1 \geq x-2 > -3$$

$$S = \{1, 3, 4\}$$

Problem 5. (6 points) Show that the function $f : \mathbb{N} \rightarrow \mathbb{Z}$ where $f(n) = (n+2)^2 - 6$ is injective but not surjective using only the definitions given in class.

Let a, b be any natural numbers. Suppose $f(a) = f(b)$. Then $(a+2)^2 - 6 = (b+2)^2 - 6$

so $|a+2| = |b+2|$. Then either

$a+2 = b+2$ hence $a = b$ or

$a+2 = -(b+2)$ but since $a \in \mathbb{N}$,

$a \geq 1$ so $a+2 \geq 3$ and cannot be

negative, thus $a+2 = -(b+2)$ is not possible.

Since $f(a) = f(b)$ implies $a = b$, f is injective.

Consider $y = -10$. Then $y \in \mathbb{Z}$ but for any

$x \in \mathbb{N}$, $x \geq 1$ so $(x+2)^2 \geq 9$

and $(x+2)^2 - 6 \geq 3$. Thus there is

no $x \in \mathbb{N}$ such that $f(x) = -10$.

Therefore f is not surjective.